

```
TC
File Edit Run Compile Project Options Debug
Line 13 Col 54 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int nr,nc,r,c,a=1;
clrscr();
printf("Enter no of rows and columns ");
scanf("%d%d",&nr,&nc);
for( r=1; r<=nr; r++ )
{
for( c=1; c<=nc; c++)
{
if(r%2==0) printf("%3d",--a); else printf("%3d",a++);_
}
printf("\n");
}
getch();
}
```

Activate Windows
Go to PC settings to activate Windows.

F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10

04:26 PM
17-Mar-25

```
TC
Enter no of rows and columns 3 4
1 2 3 4
4 3 2 1
1 2 3 4
```

Activate Windows
Go to PC settings to activate Windows.

04:27 PM
17-Mar-25

```

TC
Enter no of rows and columns 9 9
1 2 3 4 5 6 7 8 9
9 8 7 6 5 4 3 2 1
1 2 3 4 5 6 7 8 9
9 8 7 6 5 4 3 2 1
1 2 3 4 5 6 7 8 9
9 8 7 6 5 4 3 2 1
1 2 3 4 5 6 7 8 9
9 8 7 6 5 4 3 2 1
1 2 3 4 5 6 7 8 9

```

Activate Windows
Go to PC settings to activate Windows.

04:27 PM
17-Mar-25

```

a=1; 3
for( r=1; r<=nr; r++ )
{
  for( c=1; c<=nc; c++ ) 4
  {
    if(r%2==0) p(--a);
    else p(a++);
  }
  p("\n");
}

```

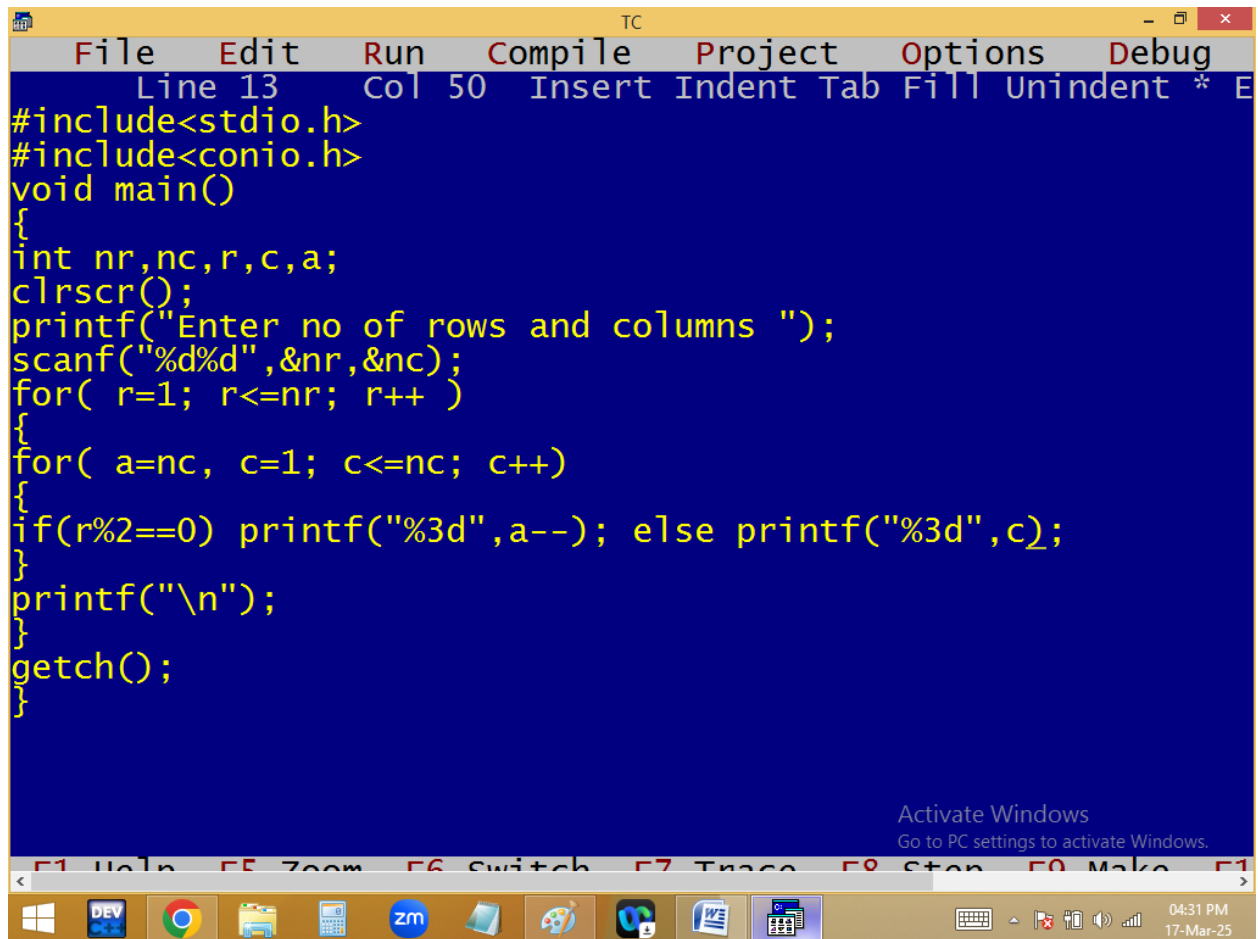
$\frac{1}{2}$ 2 3 4 5 $\frac{11}{3}$
 $\sqrt{\quad}$ $\sqrt{\quad}$ $\sqrt{\quad}$ $\sqrt{\quad}$
 4 3 2 1
 1 2 3 4

$\frac{nc}{4}$ $\frac{r}{1}$ $\frac{c}{1}$ $c = 1 \text{ to } 4$
 1
 2
 3

COLUMNS

1	1	2	3	4	} ROWS
	4	3	2	1	
3	1	2	3	4	

Method 2:



The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar at the top reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, the status bar shows "Line 13", "Col 50", and a list of keyboard shortcuts: "Insert", "Indent", "Tab", "Fill", "Unindent", "*", and "E". The main editing area has a dark blue background with yellow text. It contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int nr,nc,r,c,a;
clrscr();
printf("Enter no of rows and columns ");
scanf("%d%d",&nr,&nc);
for( r=1; r<=nr; r++ )
{
for( a=nc, c=1; c<=nc; c++)
{
if(r%2==0) printf("%3d",a--); else printf("%3d",c);
}
printf("\n");
}
getch();
}
```

At the bottom right of the IDE window, there is a watermark that says "Activate Windows" and "Go to PC settings to activate Windows." Below the IDE window is the Windows taskbar, which includes icons for Windows, DEV, Chrome, File Explorer, Calculator, Zoom, and other applications. The system clock in the bottom right corner shows "04:31 PM" and "17-Mar-25".

```

Enter no of rows and columns 5 8
1 2 3 4 5 6 7 8
8 7 6 5 4 3 2 1
1 2 3 4 5 6 7 8
8 7 6 5 4 3 2 1
1 2 3 4 5 6 7 8

```

Activate Windows
Go to PC settings to activate Windows.

```

for( r=1; r<=nr; 3r++)
{ a=nc; ✓
  for( c=1; c<=nc; 4c++)
  {
    if(r%2==0) p(a--);
    else p( c );
  }
  p("\n");
}

```

$\frac{a=nc}{2} \rightarrow 4321$
 $4 - 4321$

$\frac{nr}{3}$

$\frac{nc}{4}$

$\frac{r}{1}$

$\underline{c} = 1 \text{ to } 4$

$1 \ 2 \ 3 \ 4 - 1$

2

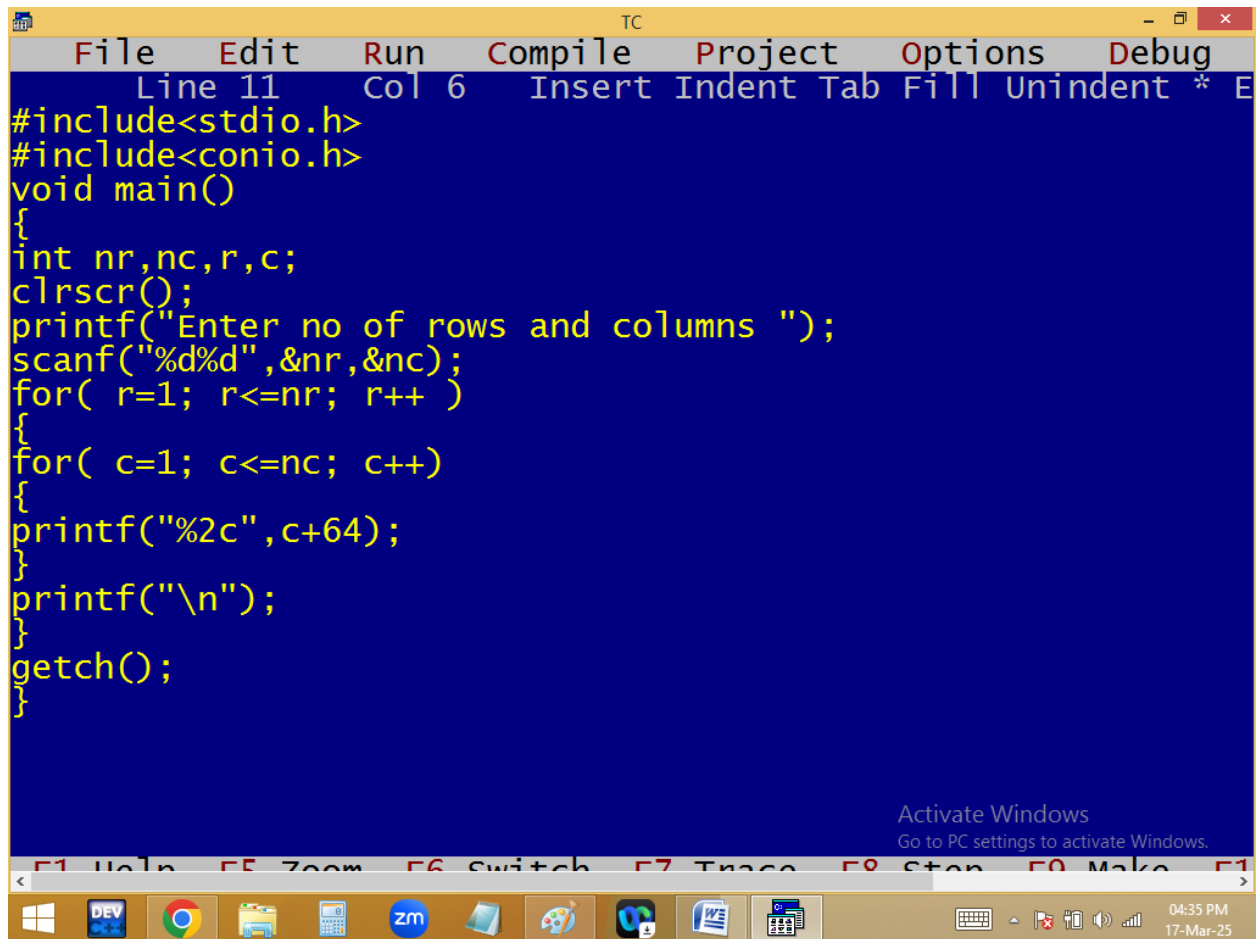
3 ✓

4

$1 \ 2 \ 3 \ 4 - 3$

COLUMNS

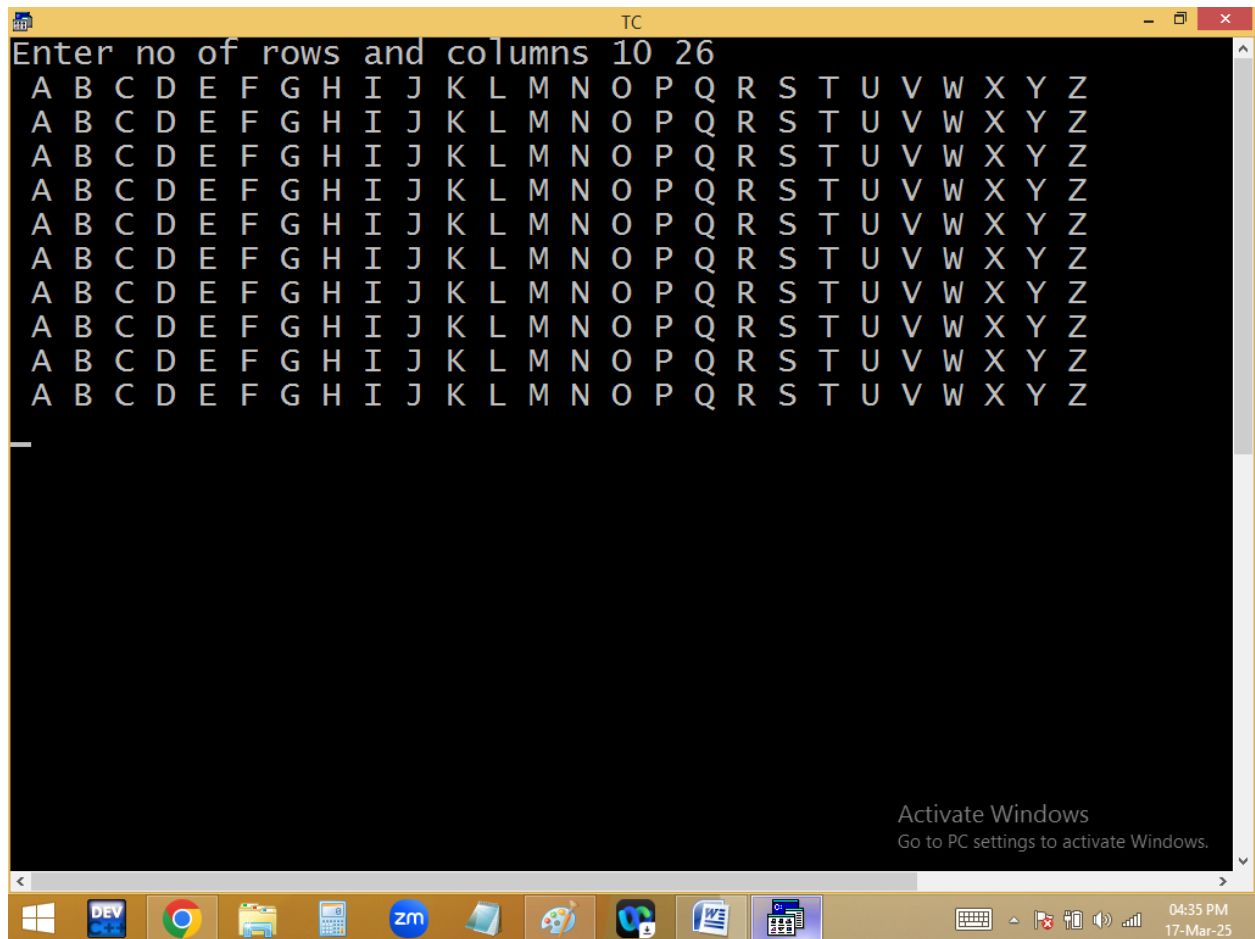
1	1	2	3	4	} ROWS
	4	3	2	1	
3	1	2	3	4	



The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, a status bar shows "Line 11", "Col 6", and a list of keyboard shortcuts: "Insert", "Indent", "Tab", "Fill", "Unindent", "*", and "E". The main editing area has a blue background and contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int nr,nc,r,c;
clrscr();
printf("Enter no of rows and columns ");
scanf("%d%d",&nr,&nc);
for( r=1; r<=nr; r++ )
{
for( c=1; c<=nc; c++)
{
printf("%2c",c+64);
}
printf("\n");
}
getch();
}
```

At the bottom right of the IDE window, there is a message: "Activate Windows. Go to PC settings to activate Windows." Below the IDE window is the Windows taskbar, which includes icons for Windows, DEV, Chrome, File Explorer, Calculator, Zoom, and several other applications. The system tray on the right shows the time as 04:35 PM and the date as 17-Mar-25.



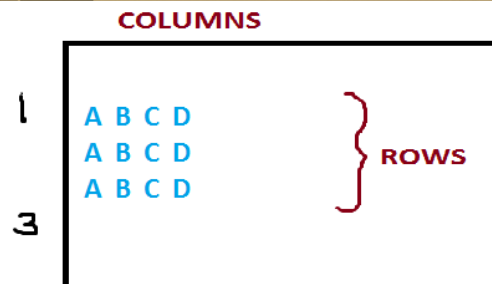
```

    3
for( r=1; r<=nr; r++ )
{ a=nc; ✓
  4
  for( c=1; c<=nc; c++ )
  {

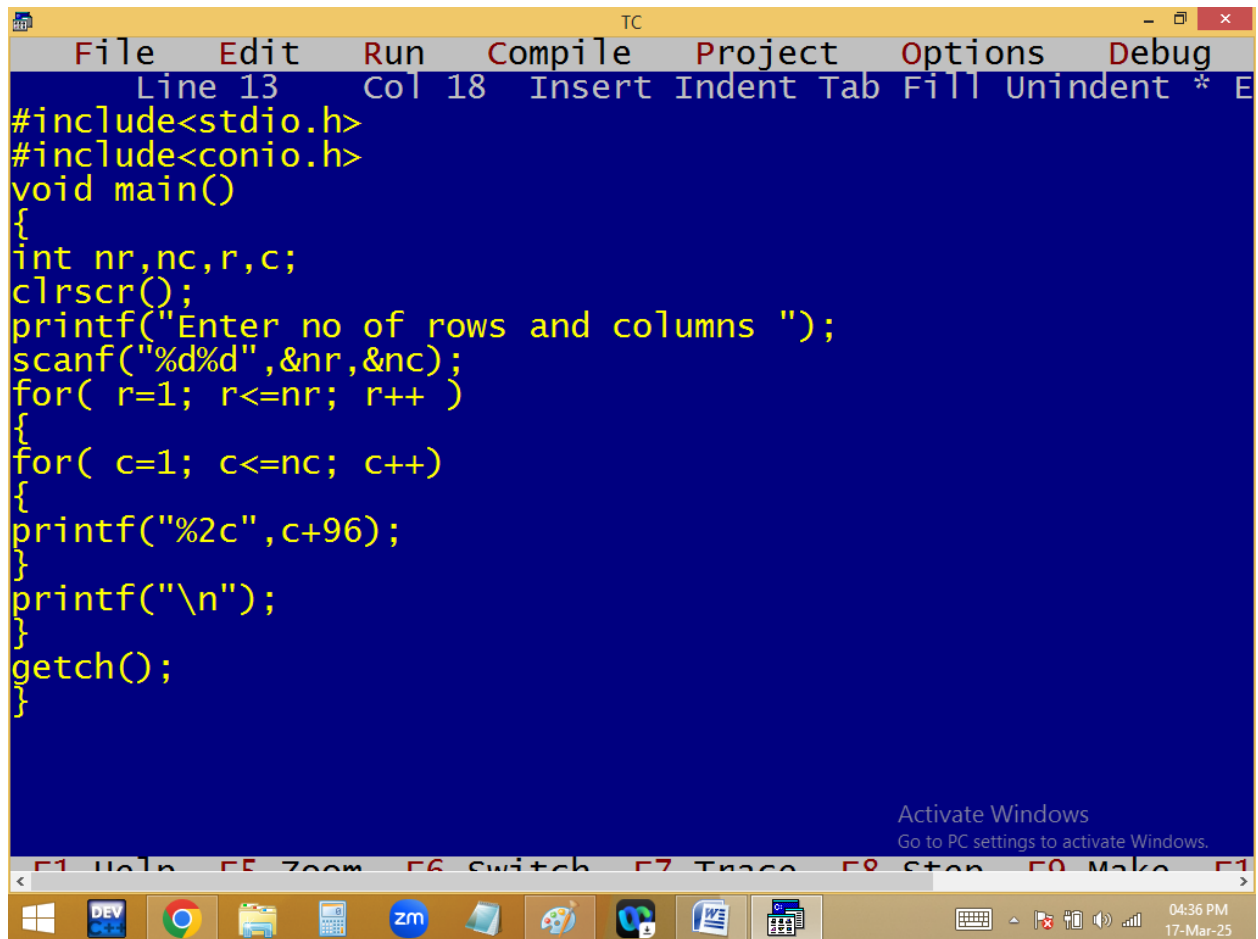
    p("%2c", c+64);

  }
  p("\n");
}

```



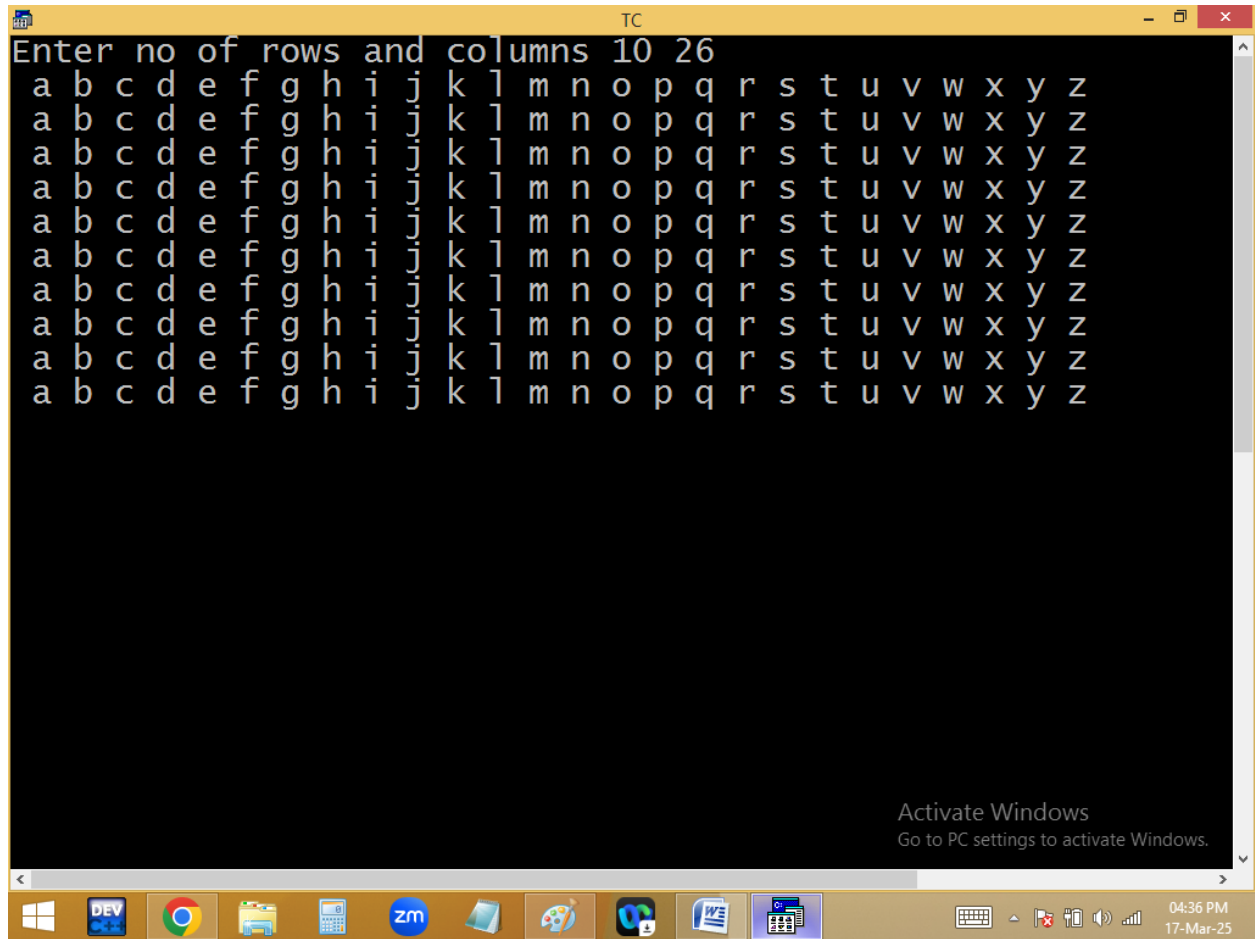
$\frac{nr}{3}$	$\frac{nc}{4}$	$\frac{r}{1}$	<u>c</u> = 1 to 4
		1	1 2 3 4 - 1
		2	
		3 ✓	1 2 3 4 - 3
		4	

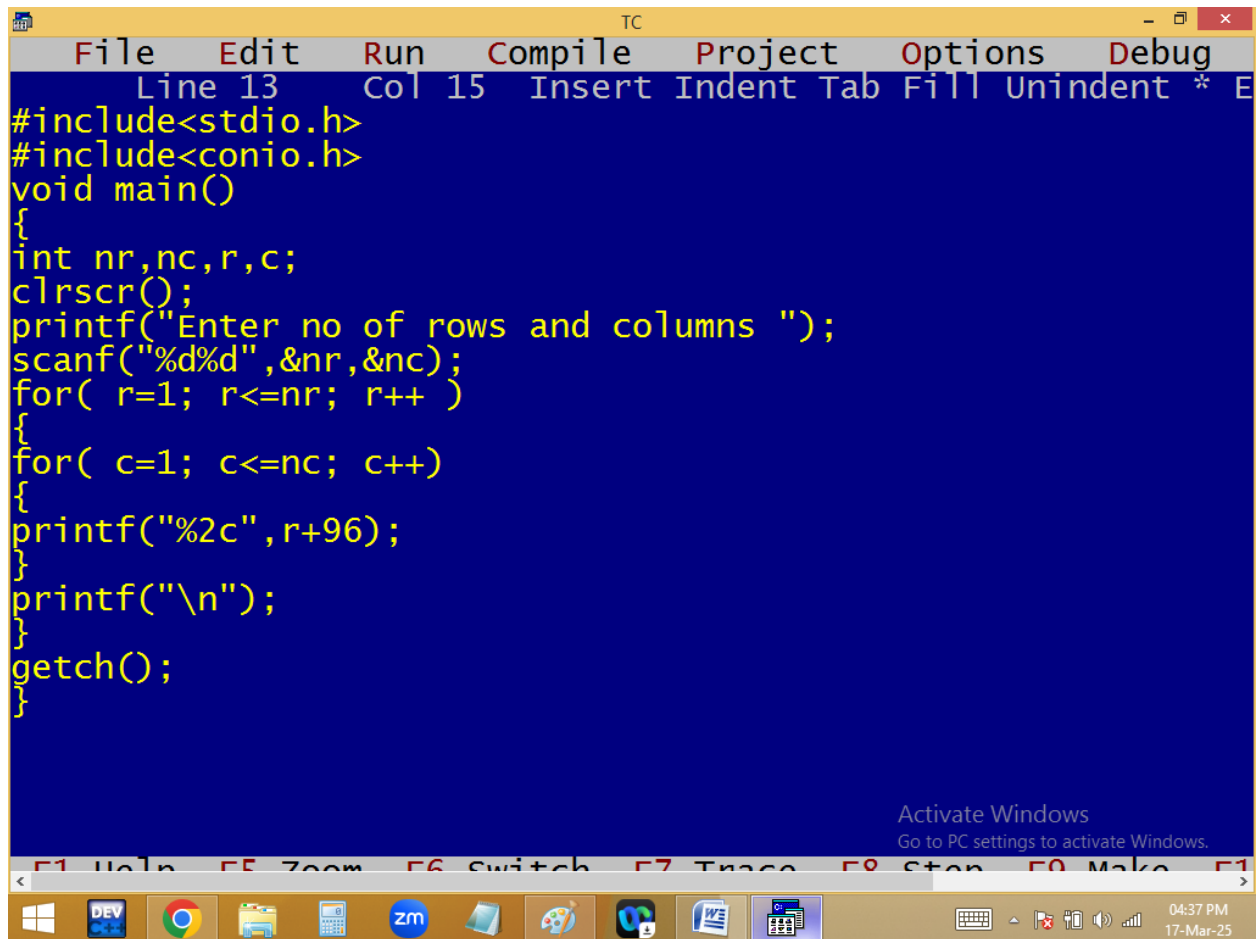


The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". The status bar at the top indicates "Line 13", "Col 18", and lists keyboard shortcuts: "Insert", "Indent", "Tab", "Fill", "Unindent", "*", and "E". The main editing area has a blue background with yellow text containing the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int nr,nc,r,c;
clrscr();
printf("Enter no of rows and columns ");
scanf("%d%d",&nr,&nc);
for( r=1; r<=nr; r++ )
{
for( c=1; c<=nc; c++)
{
printf("%2c",c+96);
}
printf("\n");
}
getch();
}
```

At the bottom right of the IDE window, there is a message: "Activate Windows. Go to PC settings to activate Windows." The Windows taskbar is visible at the bottom, showing icons for Windows, DEV, Chrome, File Explorer, Calculator, Zoom, and other applications. The system clock in the bottom right corner shows "04:36 PM" and "17-Mar-25".





The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". The status bar at the top indicates "Line 13", "Col 15", and lists keyboard shortcuts: "Insert", "Indent", "Tab", "Fill", "Unindent", "*", and "E". The main editing area has a blue background with yellow text containing the following C code:

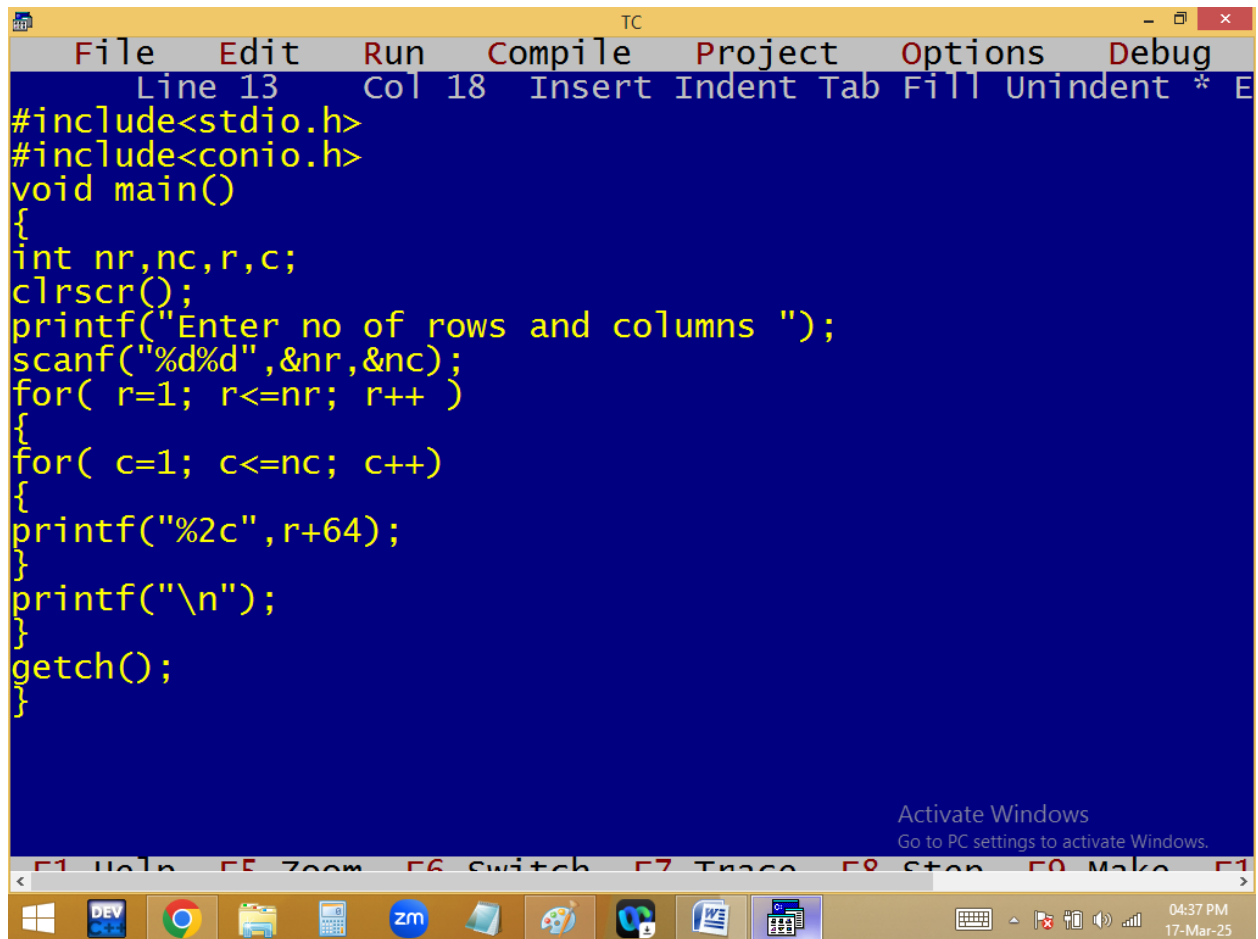
```
#include<stdio.h>
#include<conio.h>
void main()
{
int nr,nc,r,c;
clrscr();
printf("Enter no of rows and columns ");
scanf("%d%d",&nr,&nc);
for( r=1; r<=nr; r++ )
{
for( c=1; c<=nc; c++)
{
printf("%2c",r+96);
}
printf("\n");
}
getch();
}
```

At the bottom right of the IDE window, there is a message: "Activate Windows. Go to PC settings to activate Windows." The Windows taskbar is visible at the bottom, showing icons for Windows, DEV, Chrome, File Explorer, Calculator, Zoom, and other applications. The system clock in the bottom right corner shows "04:37 PM" and "17-Mar-25".

```
TC
Enter no of rows and columns 10 10
a a a a a a a a a a
b b b b b b b b b b
c c c c c c c c c c
d d d d d d d d d d
e e e e e e e e e e
f f f f f f f f f f
g g g g g g g g g g
h h h h h h h h h h
i i i i i i i i i i
j j j j j j j j j j
```

Activate Windows
Go to PC settings to activate Windows.

04:37 PM
17-Mar-25



The image shows a screenshot of a Turbo C++ (TC) IDE window. The window has a yellow title bar with the text "TC" and standard window controls. The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, a status bar shows "Line 13", "Col 18", and a list of keyboard shortcuts: "Insert", "Indent", "Tab", "Fill", "Unindent", "*", and "E". The main editing area has a blue background with yellow text. The code is as follows:

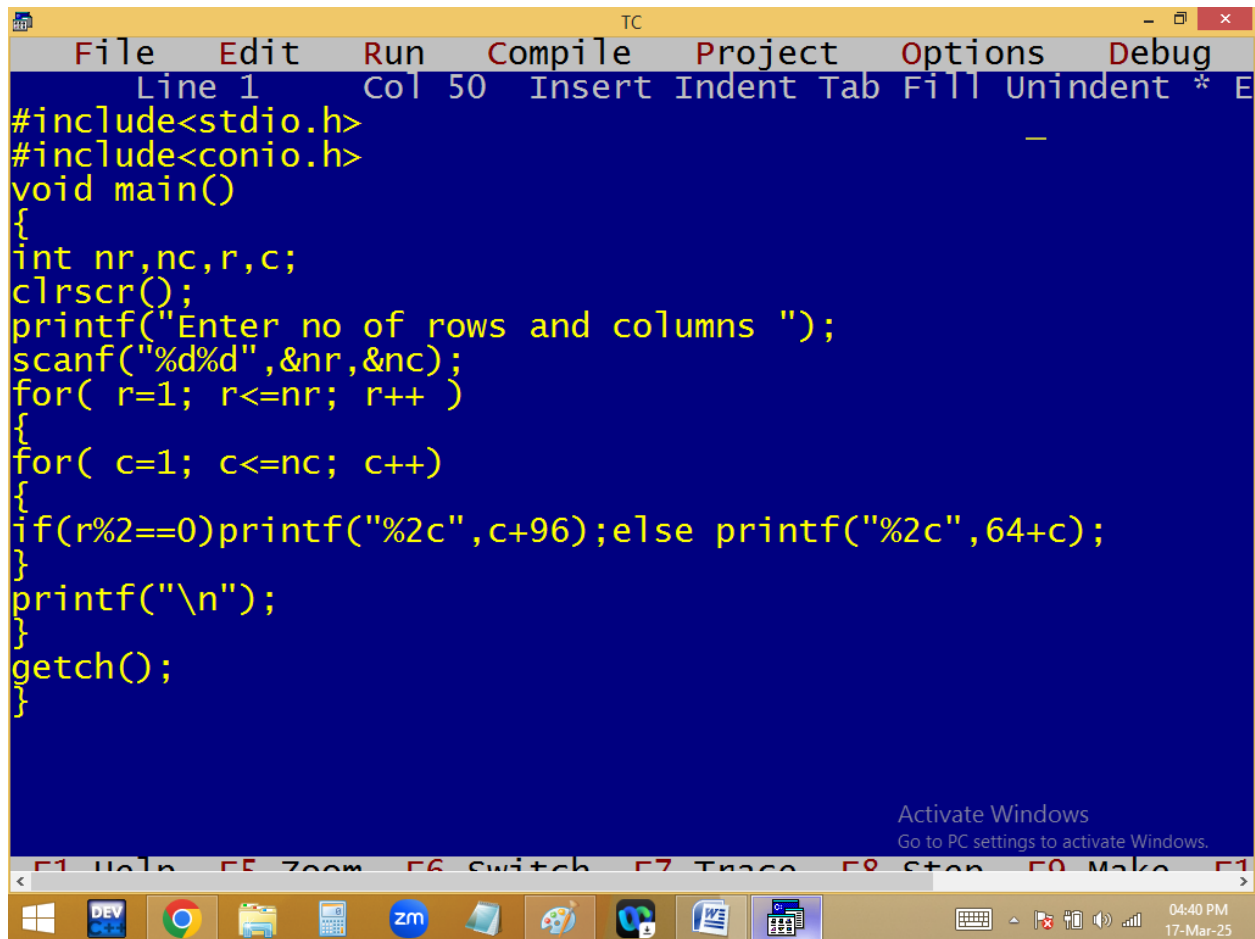
```
#include<stdio.h>
#include<conio.h>
void main()
{
int nr,nc,r,c;
clrscr();
printf("Enter no of rows and columns ");
scanf("%d%d",&nr,&nc);
for( r=1; r<=nr; r++ )
{
for( c=1; c<=nc; c++)
{
printf("%2c",r+64);
}
printf("\n");
}
getch();
}
```

At the bottom of the window, there is a text box that says "Activate Windows" and "Go to PC settings to activate Windows." Below the text box, there is a row of function key shortcuts: "F1 Help", "F5 Zoom", "F6 Switch", "F7 Trace", "F8 Stop", "F9 Make", and "F10". The Windows taskbar is visible at the bottom, showing icons for Windows, DEV, Chrome, File Explorer, Calculator, Zoom, and other applications. The system clock in the bottom right corner shows "04:37 PM" and "17-Mar-25".

```
TC
Enter no of rows and columns 10 10
A A A A A A A A A A
B B B B B B B B B B
C C C C C C C C C C
D D D D D D D D D D
E E E E E E E E E E
F F F F F F F F F F
G G G G G G G G G G
H H H H H H H H H H
I I I I I I I I I I
J J J J J J J J J J
```

Activate Windows
Go to PC settings to activate Windows.

04:37 PM
17-Mar-25



```
TC
File Edit Run Compile Project Options Debug
Line 1 Col 50 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int nr,nc,r,c;
clrscr();
printf("Enter no of rows and columns ");
scanf("%d%d",&nr,&nc);
for( r=1; r<=nr; r++ )
{
for( c=1; c<=nc; c++)
{
if(r%2==0)printf("%2c",c+96);else printf("%2c",64+c);
}
printf("\n");
}
getch();
}
```

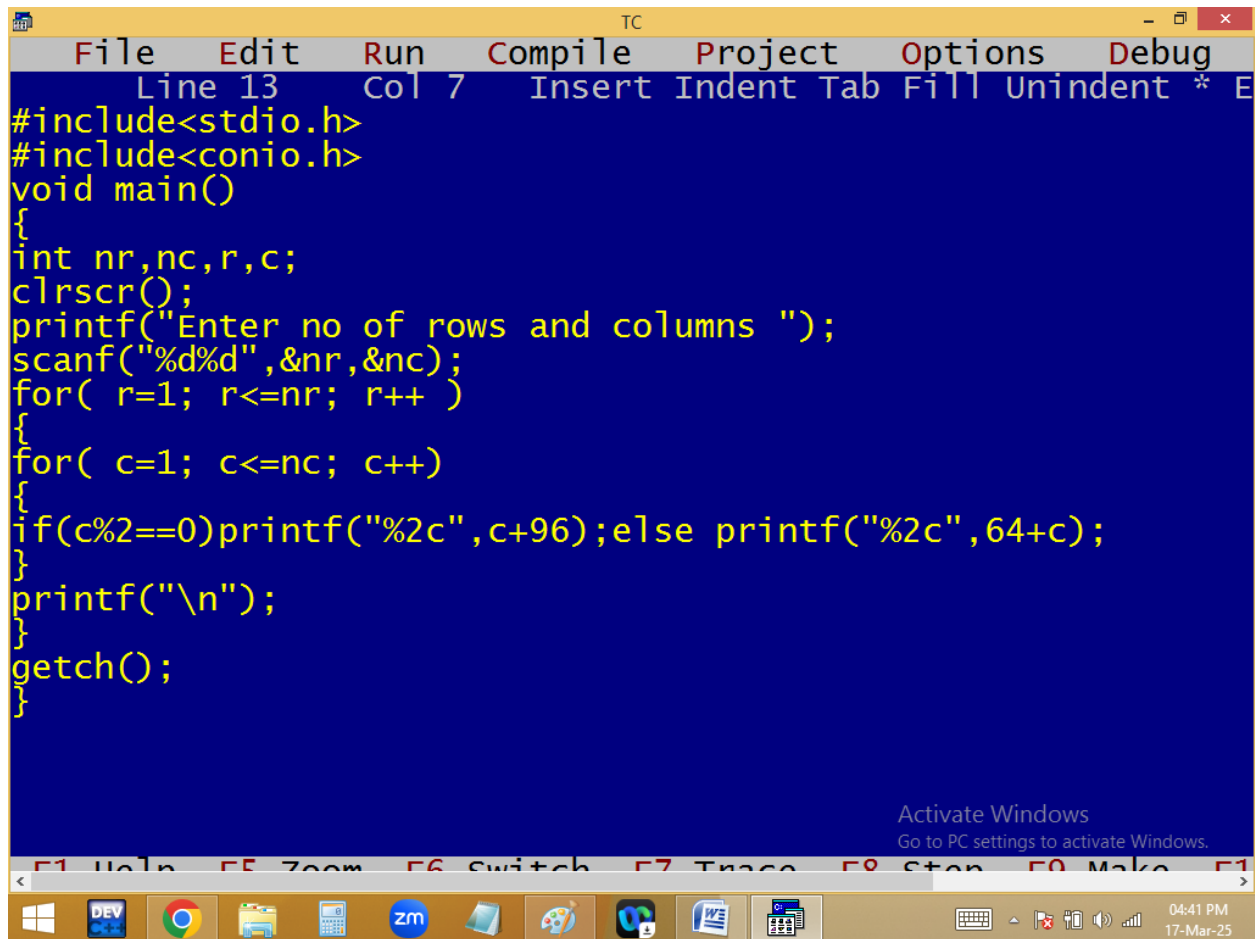
Activate Windows
Go to PC settings to activate Windows.

F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10

04:40 PM
17-Mar-25

```
TC
Enter no of rows and columns 10 26
A B C D E F G H I J K L M N O P Q R S T U V W X Y Z
a b c d e f g h i j k l m n o p q r s t u v w x y z
A B C D E F G H I J K L M N O P Q R S T U V W X Y Z
a b c d e f g h i j k l m n o p q r s t u v w x y z
A B C D E F G H I J K L M N O P Q R S T U V W X Y Z
a b c d e f g h i j k l m n o p q r s t u v w x y z
A B C D E F G H I J K L M N O P Q R S T U V W X Y Z
a b c d e f g h i j k l m n o p q r s t u v w x y z
A B C D E F G H I J K L M N O P Q R S T U V W X Y Z
a b c d e f g h i j k l m n o p q r s t u v w x y z

Activate Windows
Go to PC settings to activate Windows.
```

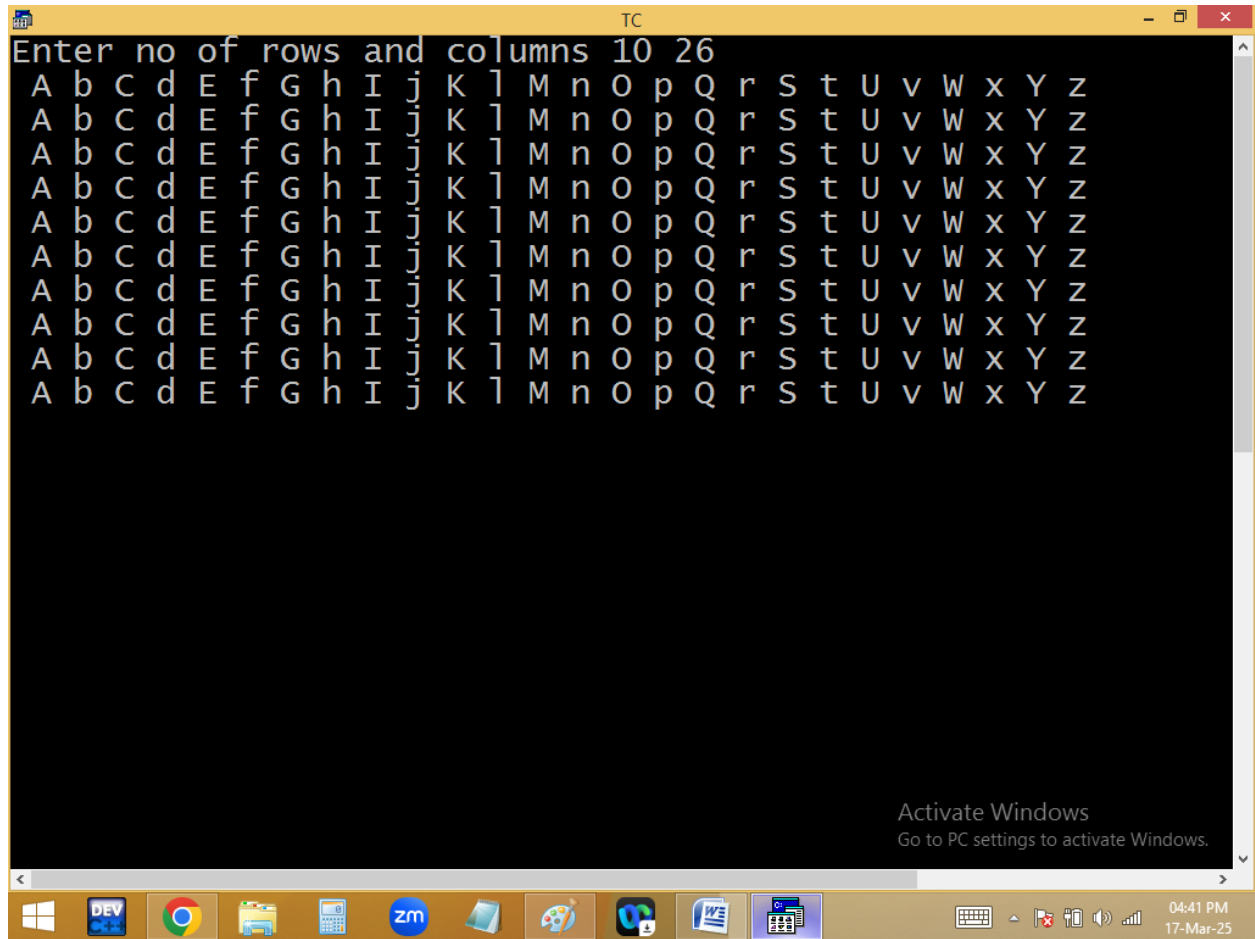


```
TC
File Edit Run Compile Project Options Debug
Line 13 Col 7 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int nr,nc,r,c;
clrscr();
printf("Enter no of rows and columns ");
scanf("%d%d",&nr,&nc);
for( r=1; r<=nr; r++ )
{
for( c=1; c<=nc; c++)
{
if(c%2==0)printf("%2c",c+96);else printf("%2c",64+c);
}
printf("\n");
}
getch();
}
```

Activate Windows
Go to PC settings to activate Windows.

F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10

04:41 PM
17-Mar-25



```
TC
File Edit Run Compile Project Options Debug
Line 1 Col 24 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int nr,nc,r,c; char L='a', U='A';
clrscr();
printf("Enter no of rows and columns ");
scanf("%d%d",&nr,&nc);
for( r=1; r<=nr; r++ )
{
for( c=1; c<=nc; c++)
{
if(r%2==0)printf("%2c",L);else printf("%2c",U);
}
printf("\n"); if(r%2==0)L++;else U++;
}
getch();
}
```

Activate Windows
Go to PC settings to activate Windows.

F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10

05:04 PM
17-Mar-25

```
TC
Enter no of rows and columns 14 14
A A A A A A A A A A A A A A A
a a a a a a a a a a a a a a a
B B B B B B B B B B B B B B B
b b b b b b b b b b b b b b b
C C C C C C C C C C C C C C C
c c c c c c c c c c c c c c c
D D D D D D D D D D D D D D D
d d d d d d d d d d d d d d d
E E E E E E E E E E E E E E E
e e e e e e e e e e e e e e e
F F F F F F F F F F F F F F F
f f f f f f f f f f f f f f f
G G G G G G G G G G G G G G G
g g g g g g g g g g g g g g g

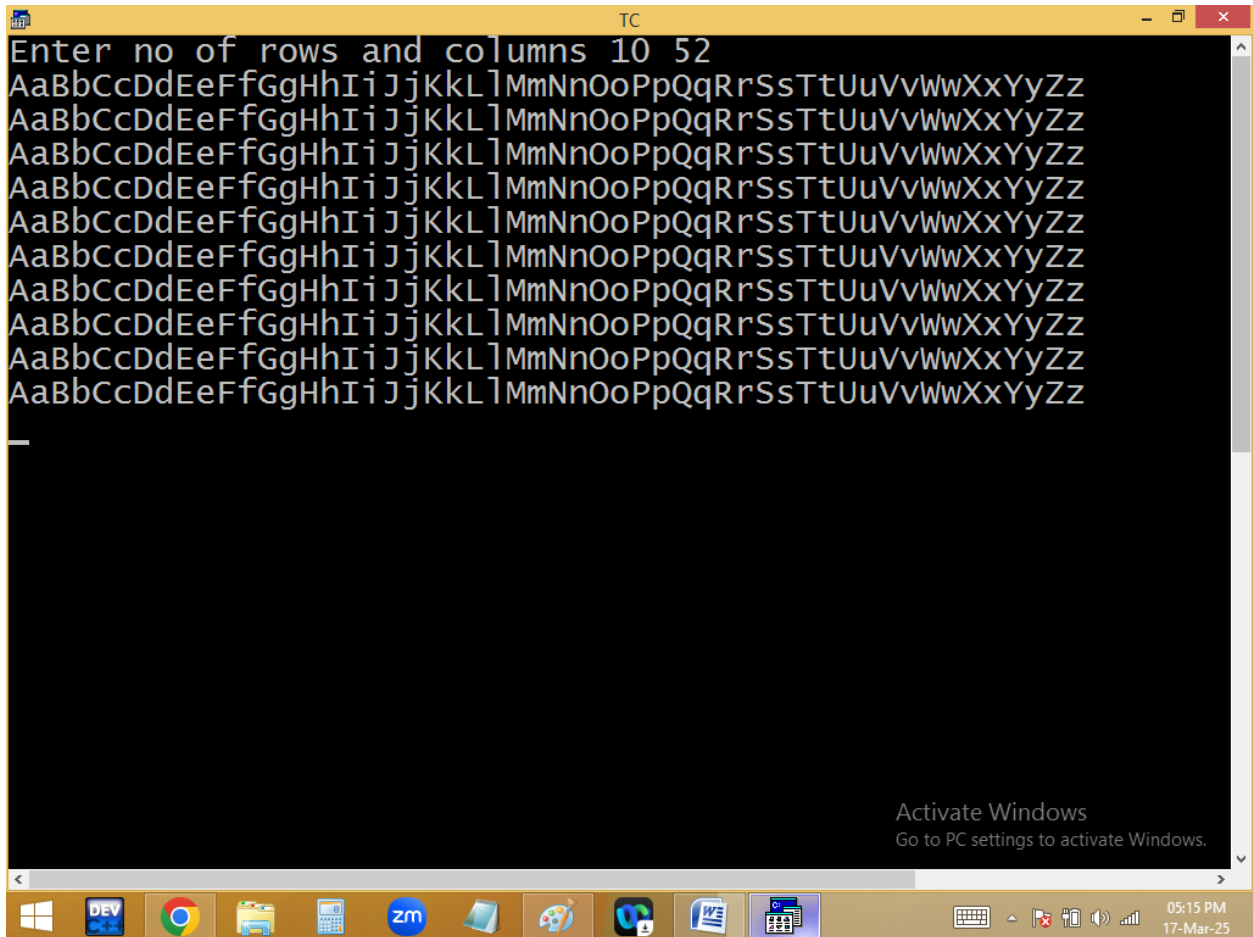
Activate Windows
Go to PC settings to activate Windows.
```

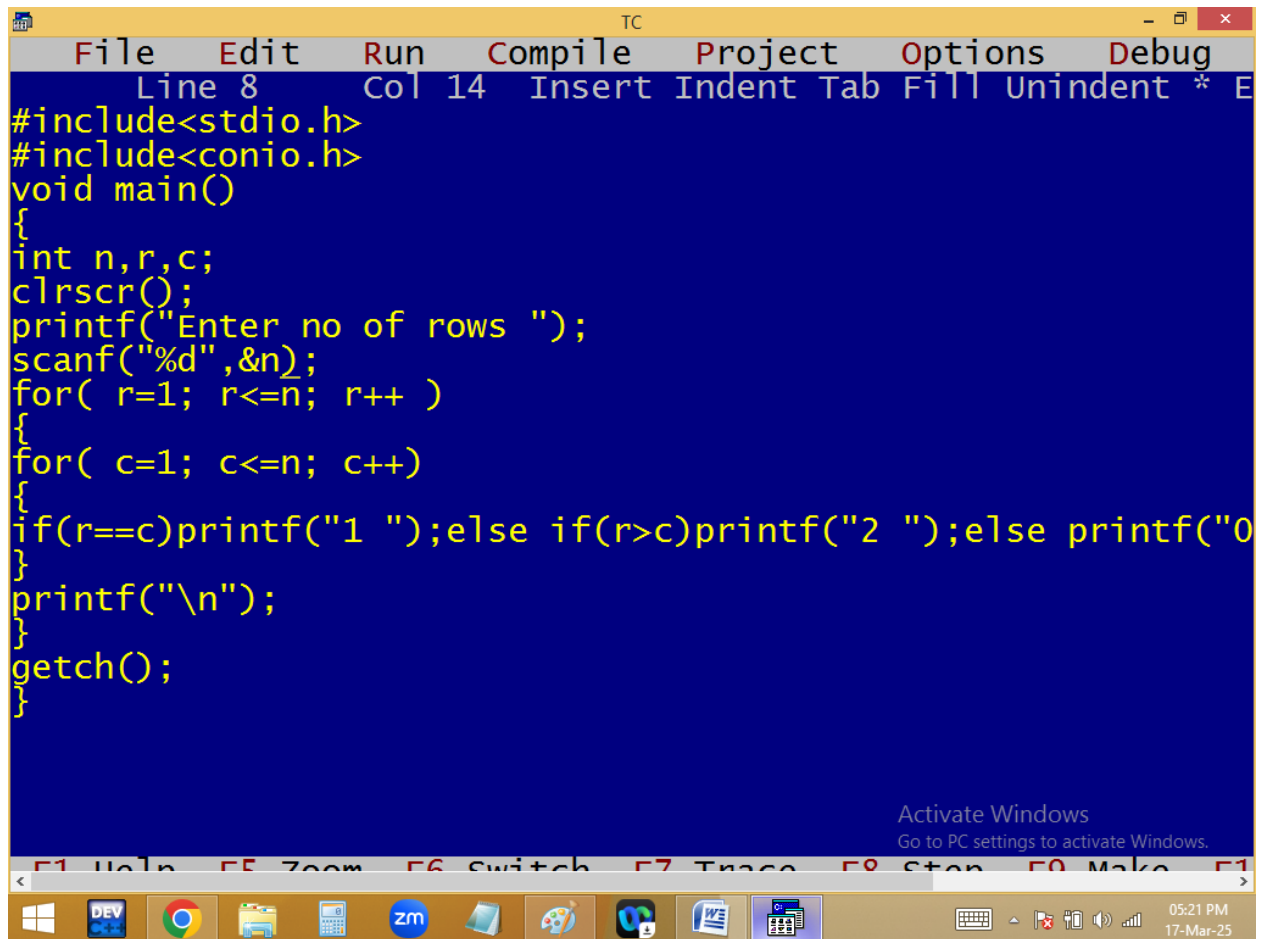
```
TC
File Edit Run Compile Project Options Debug
Line 13 Col 42 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int nr,nc,r,c;
clrscr();
printf("Enter no of rows and columns ");
scanf("%d%d",&nr,&nc);
for( r=1; r<=nr; r++ )
{char L= a , U= A ;
for( c=1; c<=nc; c++)
{
if(c%2==0)printf("%c",L++);else printf("%c",U++);
}
printf("\n");
}
getch();
}
```

Activate Windows
Go to PC settings to activate Windows.

F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10

05:15 PM
17-Mar-25



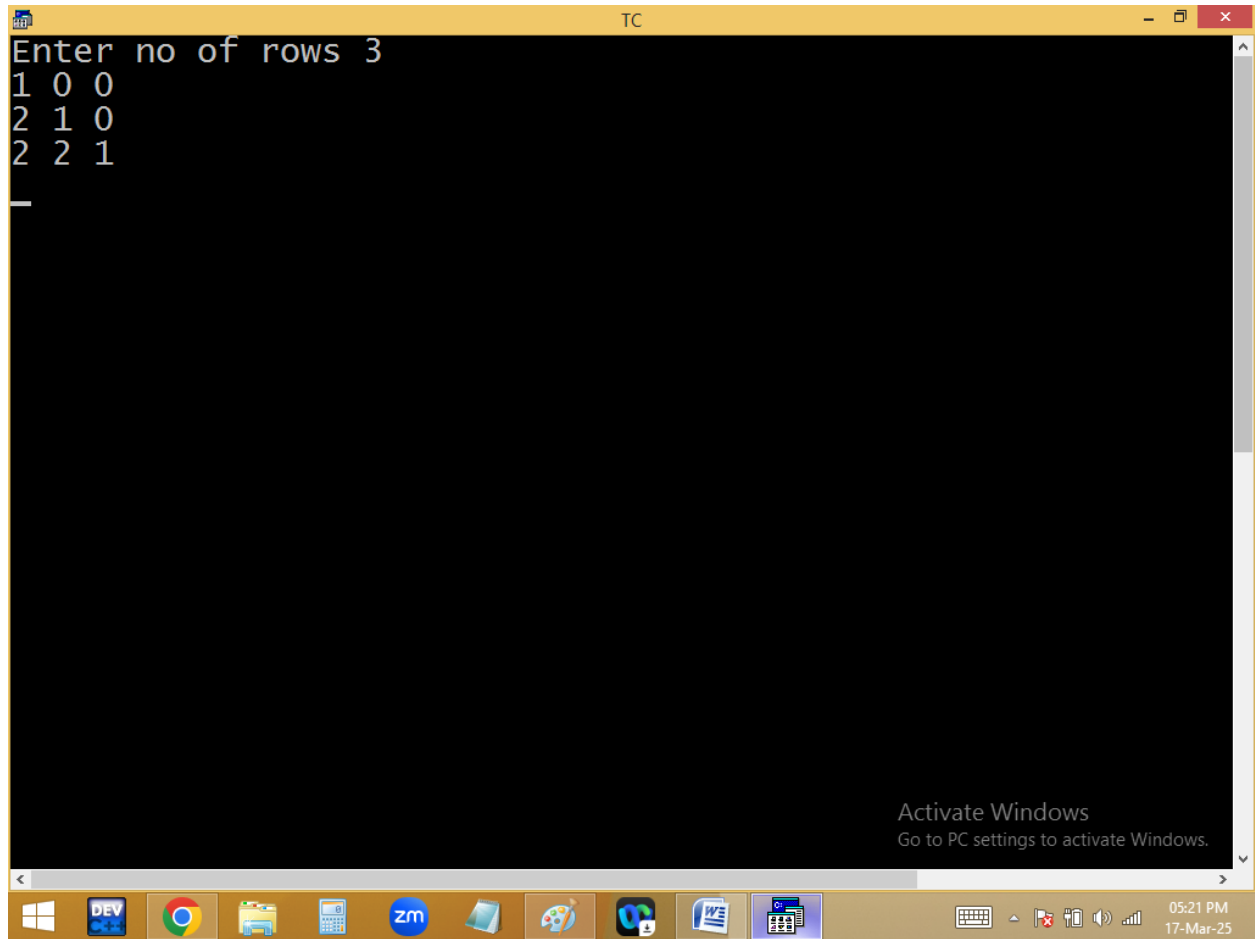


```
TC
File Edit Run Compile Project Options Debug
Line 8 Col 14 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int n,r,c;
clrscr();
printf("Enter no of rows ");
scanf("%d",&n);
for( r=1; r<=n; r++ )
{
for( c=1; c<=n; c++)
{
if(r==c)printf("1 ");else if(r>c)printf("2 ");else printf("0
}
printf("\n");
}
getch();
}
```

Activate Windows
Go to PC settings to activate Windows.

F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10

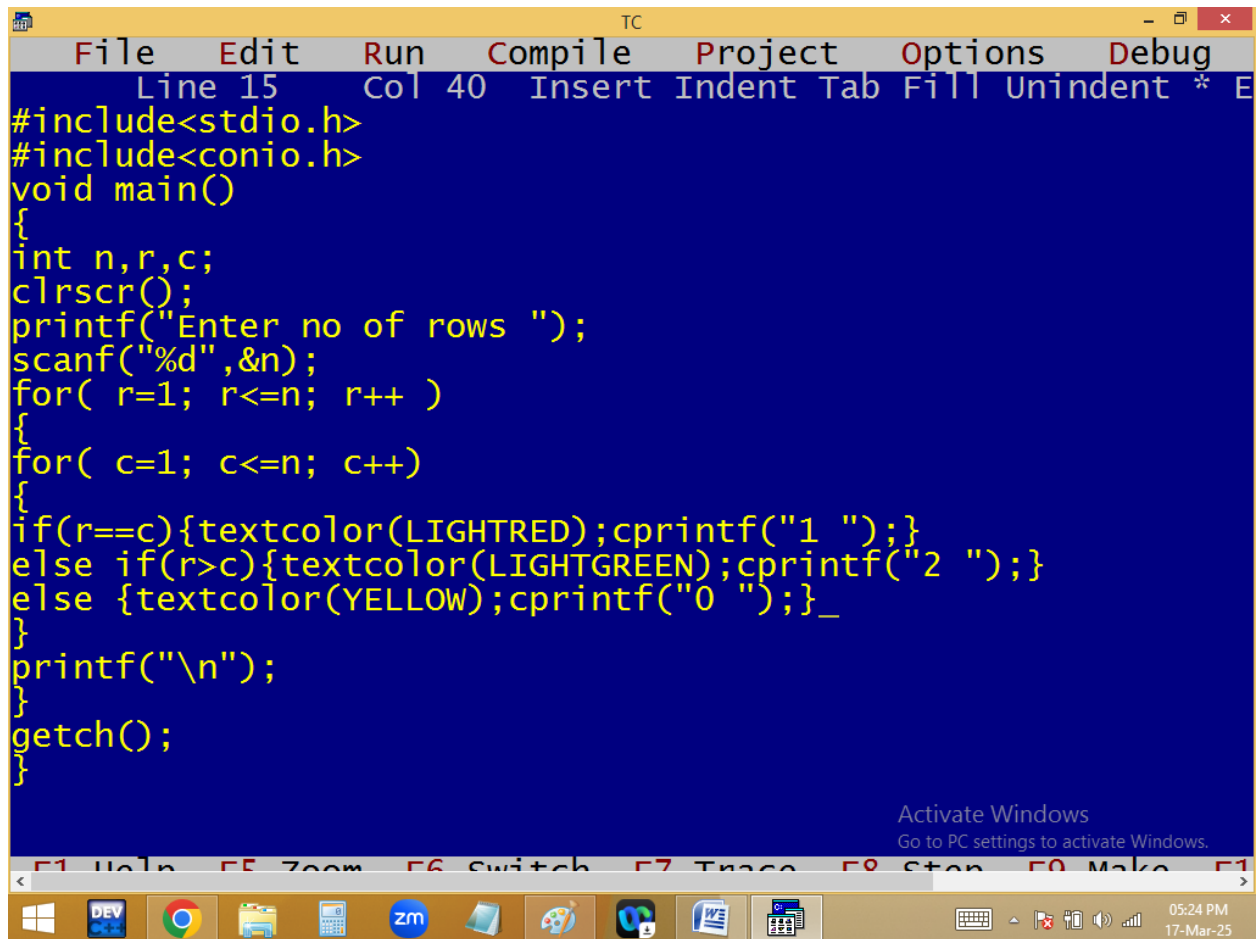
Windows taskbar: DEV, Chrome, File Explorer, Calculator, Zoom, Paint, VS Code, Word, Excel, System tray (05:21 PM, 17-Mar-25)



```
TC
Enter no of rows 10
1 0 0 0 0 0 0 0 0 0
2 1 0 0 0 0 0 0 0 0
2 2 1 0 0 0 0 0 0 0
2 2 2 1 0 0 0 0 0 0
2 2 2 2 1 0 0 0 0 0
2 2 2 2 2 1 0 0 0 0
2 2 2 2 2 2 1 0 0 0
2 2 2 2 2 2 2 1 0 0
2 2 2 2 2 2 2 2 1 0
2 2 2 2 2 2 2 2 2 1
```

Activate Windows
Go to PC settings to activate Windows.

05:22 PM
17-Mar-25



The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes File, Edit, Run, Compile, Project, Options, and Debug. The status bar at the top indicates "Line 15 Col 40" and lists keyboard shortcuts: Insert, Indent, Tab, Fill, Unindent, *, and E. The main editing area has a blue background and contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int n,r,c;
clrscr();
printf("Enter no of rows ");
scanf("%d",&n);
for( r=1; r<=n; r++ )
{
for( c=1; c<=n; c++)
{
if(r==c){textcolor(LIGHTRED);cprintf("1 ");}
else if(r>c){textcolor(LIGHTGREEN);cprintf("2 ");}
else {textcolor(YELLOW);cprintf("0 ");}_
}
printf("\n");
}
getch();
}
```

At the bottom right of the IDE, there is a message: "Activate Windows Go to PC settings to activate Windows." The Windows taskbar is visible at the bottom, showing icons for Windows, DEV, Chrome, File Explorer, Calculator, Zoom, and other applications. The system clock in the bottom right corner shows "05:24 PM 17-Mar-25".

```
TC
Enter no of rows 10
1 0 0 0 0 0 0 0 0 0
2 1 0 0 0 0 0 0 0 0
2 2 1 0 0 0 0 0 0 0
2 2 2 1 0 0 0 0 0 0
2 2 2 2 1 0 0 0 0 0
2 2 2 2 2 1 0 0 0 0
2 2 2 2 2 2 1 0 0 0
2 2 2 2 2 2 2 1 0 0
2 2 2 2 2 2 2 2 1 0
2 2 2 2 2 2 2 2 2 1
```

Activate Windows
Go to PC settings to activate Windows.

05:25 PM
17-Mar-25

1 0 0

1,1

1,2

1,3

2 1 0

2,1

2,2

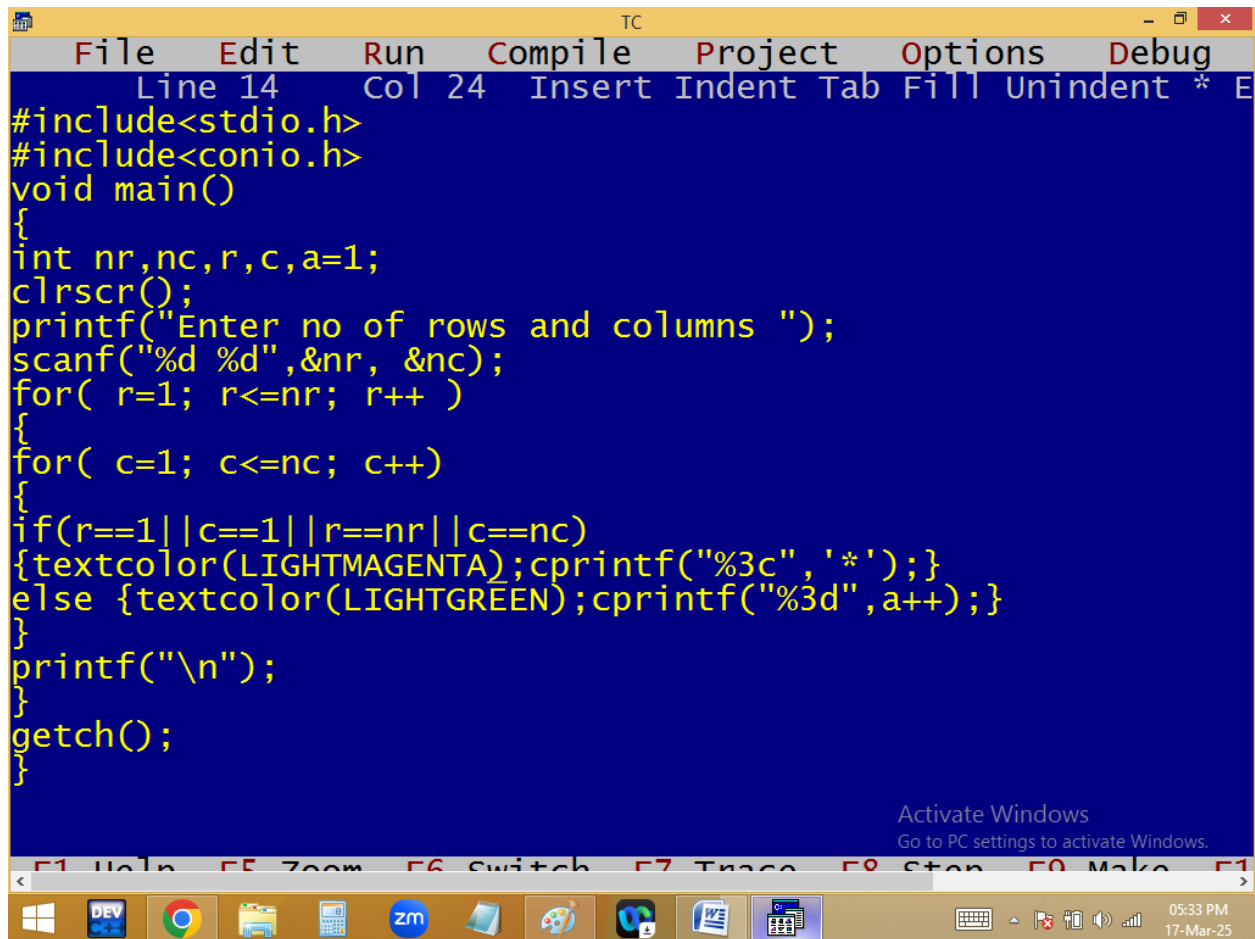
2,3

2 2 1

3,1

3,2

3,3



```
TC
File Edit Run Compile Project Options Debug
Line 14 Col 24 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int nr,nc,r,c,a=1;
clrscr();
printf("Enter no of rows and columns ");
scanf("%d %d",&nr, &nc);
for( r=1; r<=nr; r++ )
{
for( c=1; c<=nc; c++)
{
if(r==1||c==1||r==nr||c==nc)
{textcolor(LIGHTMAGENTA);cprintf("%3c",'*');}
else {textcolor(LIGHTGREEN);cprintf("%3d",a++);}
}
printf("\n");
}
getch();
}
```

Activate Windows
Go to PC settings to activate Windows.

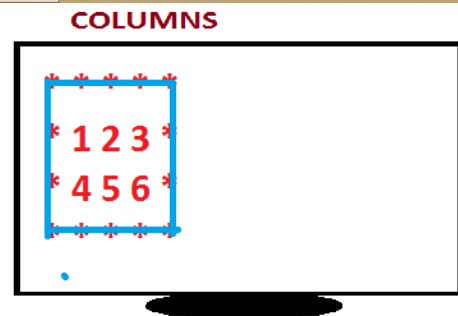
F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10

Windows taskbar: DEV, Chrome, File Explorer, Calculator, Zoom, Paint, VS Code, Word, Excel, System tray (05:33 PM, 17-Mar-25)

```
TC
Enter no of rows and columns 10 10
* * * * *
* 1 2 3 4 5 6 7 8 *
* 9 10 11 12 13 14 15 16 *
* 17 18 19 20 21 22 23 24 *
* 25 26 27 28 29 30 31 32 *
* 33 34 35 36 37 38 39 40 *
* 41 42 43 44 45 46 47 48 *
* 49 50 51 52 53 54 55 56 *
* 57 58 59 60 61 62 63 64 *
* * * * *

Activate Windows
Go to PC settings to activate Windows.
```

```
a=1;
if( r==1 || c==1 || c==nc || r==nr) p(*);
else p(a++);
```



```
if(r==c)p(1);
else if(r>c) p(2);
else p(0);
```

Home work:

n	1 1 1 1
	1 1 2 2
4	1 3 3 3
	4 4 4 4

		rows=4	rows=4	row3	4rows
1 2 3 4	4 3 2 1	1	10	6	0
1 2 3	3 2 1	2 3	9 8	5 4	1 1
1 2	2 1	4 5 6	7 6 5	3 2 1	2 3 5
1	1	7 8 9 10	4 3 2 1		8 13 21 34

rows=4

1

2 4

3 5 7

6 8 10 12

